

Eighth International Conference on Futuristic Trends in Networks and Computing Technologies (FTNCT08)

Performance Benchmarking of NFT-Enabled Hyperledger Fabric Network for Automotive Supply Chains in Smart Cities

Archana Kurde^a, Sushil Kumar Singh^{b,*}

^{a,b}*Department of Computer Engineering, Marwadi University, Rajkot, India;*

Abstract

The swift expansion of IoT devices in smart cities demands decentralized and open systems of attentive exchange of assets in automotive supply chains. Nevertheless, the majority of the available blockchain-based solutions are focused on traceability and ignore scalability, conditional payment automation, and real-time IoT verification. To overcome those pitfalls, the research proposes a Blockchain-based framework implemented on Hyperledger Fabric, incorporating Non-Fungible Tokens, and escrow-based smart contracts, to facilitate verifiable, automated vehicle transactions. The payment is conditionally released, and the vehicle is represented as a discrete NFT that undergoes authenticated release under Fabric Certificate Authority with escrow verification. Sub-millisecond latency (0.0003 s), constant throughput, and minimal computational cost experimentally verify the effectiveness of the framework in terms of its efficiency, privacy, and scalability in the efficient and autonomous exchange of assets in next-generation smart cities.

© 2026 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the Eighth International Conference on Futuristic Trends in Networks and Computing Technologies (FTNCT08).

Keywords: Permissioned Blockchain; Distributed Ledger; Digitized Asset

1. Introduction

Smart cities are growing rapidly due to the deployment of a large number of IoT devices within the subsystems, namely Smart Hospitals, Smart Buildings, Smart Supply Chain, Smart Retail, and many more. For sustainable development and optimized resource utilization, seamless data exchange and automated decision-making are needed. The convergence of diversified subsystems collectively contributing to the sustainable development of the individuals residing in them encounters several challenges in interoperability of heterogeneous systems, data privacy, scalability, and trust management. These challenges need to be addressed critically for providing an improved, non-interrupted,

Corresponding Author Email: sushilkumar.singh@marwadieducation.edu.in

and safe experience of the infrastructure developed within the Smart city. Conventional Supply Chains rely on a central authority for verification, authentication, and data storage, which causes the issue of single-point failure. Such centralized systems are inefficient for privacy preservation, transparency and are prone to asset counterfeiting.

Blockchain Technology has tremendous potential to provide efficient and feasible solutions for these challenges in the by improving traceability, decentralized identity management, and transactions [1]. The public Blockchain networks lack data privacy due to uncontrolled participation and governance, vulnerable to network-level attacks [2]. Over it, Hyperledger Fabric is a permissioned Blockchain framework that supports decentralized asset management through non-fungible token (NFT) and programmable smart contracts that are triggered by events forming the workflow of individual and asset authentication. These capabilities make them suitable for industrial ecosystems that demand authentication, automated asset traceability, and regulated access control. Supply chain, being a constituent part of a Smart city can leverage these operational abilities to address the challenges encountered within the network. This research introduces a Blockchain-enabled NFT-based framework [3,4] tailored for the smart automotive supply chain with enhanced transparency and traceability with digitized asset creation and management.

The unifying objective of this study is designing and implementing a transparent, traceable, and trustworthy Blockchain-enabled framework for smart automotive supply chains embedded in a Smart city. The key contributions of this research include a unique digital asset management module utilizing NFT for managing the transfer of asset among multiple stakeholders. Additionally, a conditional escrow service ensures the fund is released only after successful transfer of ownership to the intended receiver. The performance of the proposed framework has been monitored using Prometheus to measure throughput, latency, and block height during the entire workflow, beginning from asset creation till ownership transfer to the end customer. The rest of this paper is organized as follows: Related Work, Proposed Methodology, Results, and Conclusion.

2. Related Work

Recent research on blockchain-enabled supply chains has primarily focused on enhancing traceability, authenticity, and trust management in a distributed environment. Alsadi et al. [5] proposed TruCert, a blockchain-based certification system for the automotive sector using Ethereum, which ensures product provenance but lacks IoT integration and conditional payment automation. The authentication relies on a central authority for certification using a public key, which exposes the system to a single point of failure and assumption of an uncompromised key used for authentication. Chen et al. [6] developed a Hyperledger Fabric-based traceability model for the tea supply chain using off-chain IPFS storage, offering scalability but no on-chain automation with a missing monitoring framework to measure performance matrices. Iftexhar et al. [7] implemented a public Ethereum solution for food safety assurance through IoT data logging, although with higher latency and limited privacy control. The system lacks telemetry performance analysis, a multi-organization architecture to visualize performance during peak requests, and no ownership concept to ensure the safety of transit within multiple stakeholders. Li et al. [8] managed identity using a public-private key with data integrity using SHA-256 with simple access control rules to assure access by authorized individuals. Incorporation of role-based authentication could enhance the trust among stakeholders.

Fabric-TeaChain [9] utilized an immutable ledger to record the events involved in the entire process of tea harvesting, with QR codes linking physical Tea packages to provide consumer transparency, product provenance, and anti-counterfeit verification, despite real-time performance monitoring, and a financial layer is missing. PrivChain [10] employs zero-knowledge proofs within Hyperledger to enhance privacy and data confidentiality, whereas DeTRM [11] introduces a decentralized trust and reputation model for blockchain-based supply chains. However, both frameworks overlook real-time IoT-based validation and escrow automation. Patil et al. [12] introduce an energy-conscious blockchain architecture for sustainable industrial IoT with an efficient IoT-Blockchain interaction. It provides a conceptual framework with a lack of implementation and testing on real-time data, along with a missing threat model or security proofs. Qureshi et al. [13] provide a detailed survey of green blockchain techniques concentrating on carbon footprint reduction and sustainable chain design enabled with an eco-friendly consensus mechanism. Also, being a survey only, restrains the work from a practical implementation framework proposing a green consensus mechanism for sustainable Blockchain-based solutions.

Most existing Blockchain-based solutions leverage public Blockchain platforms that lack the scalability much needed for industrial deployment and privacy control. The proposed framework leverages Hyperledger Fabric as a

permissioned Blockchain platform that utilizes Non-Fungible Tokens (NFTs) for unique digital asset representation of a car [14–17]. The framework automates conditional transactions by integrating smart contracts and escrow mechanisms that ensure transfer and payment only when predefined conditions are met [18,19]. This approach provides transparency to the entire process of asset transfer, beginning from asset creation till its ownership transfer to the end customer, and minimizes human intervention apart from mitigating the risk of fraud and disputes among stakeholders.

The suggested framework will allow a single framework to manage and provide automotive assets on Hyperledger Fabric in the form of NFT-based digital identities and settle settlements transparently and conditionally with the help of escrow smart contracts. Using IoT data streams, it is used to provide real-time monitoring and confirmation of asset positioning [20]. The scalability, latency, and throughput of the system were tested with the help of Prometheus, which is an integrated tool to measure metrics [21–23] that showed improved efficiency, confidence, and sustainability in next-generation automotive supply chains with a smooth incorporation of Blockchain to IoT.

3. Proposed Methodology

The proposed framework creates a transparent and secure smart automotive supply chain that consists of three major layers, as shown in Figure 1. The top layer consists of stakeholders, namely manufacturer, dealer and end customer, who communicate with the system to mint vehicle NFTs, initiate escrow, and complete ownership transfer. The middle layer has Org1 responsible for NFT creation and management, Org2 manages escrow service and asset transfer operation upon confirmation that the required conditions are met, orderer provides Raft-based consensus for block formation and broadcasting. The bottom layer has IoT sensors and Prometheus monitoring to supply real-time telemetry to measure the performance of the system. To apply the proposed delivery model, Hyperledger Fabric v2.x was used to grant the participants the opportunity to exchange information safely and permissioned [24]. The network consisted of three organizations with Org1 (Manufacturer) and Org2 (Dealer), and Org3 the Order-er Organization, one peer, and a Certificate Authority. The cryptographic identities were dynamically managed under Fabric CA in registration and enrolment [25–29].

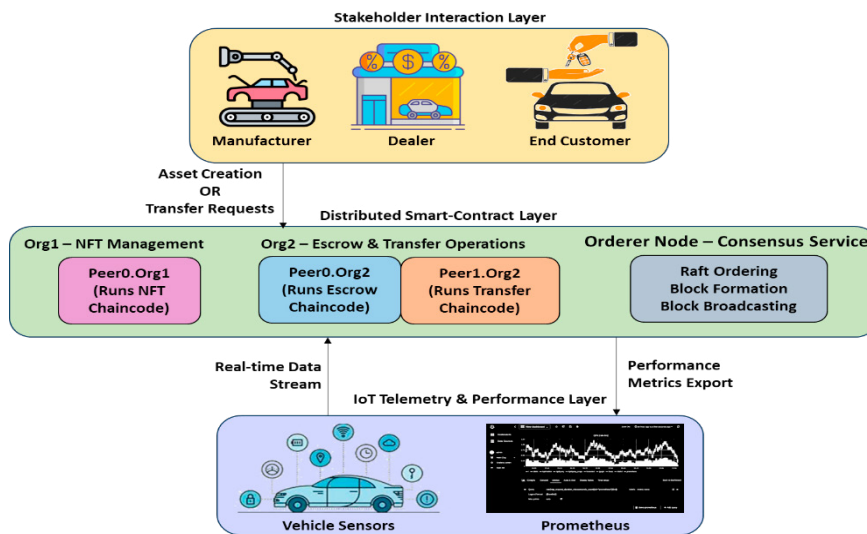


Fig. 1. Architecture of the Proposed Framework for Smart Automotive Supply Chain

Algorithm 1. NFT Creation Process

Input: Asset details A, Owner O, SmartContract C

Output: Unique NFT tokenID

1. function CREATE_NFT (A, O)
2. Verify authenticity(A)
3. metadata $\leftarrow$ generateMetadata(A)

4. $\text{tokenID} \leftarrow \text{hash}(\text{metadata} \parallel \text{timestamp})$
5. $\text{C.mint}(\text{tokenID}, \text{O}, \text{metadata})$
6. Record transaction (tokenID, O)
7. return tokenID

NFTs, initiation of escrows, validation with the help of IoT, and conditional transfer of ownership, and the orderer provided the consensus and impeccability of the ledger were created using Go Smart contracts. Artifacts, identities enrolled, and mychannel were created and joined using configtxgen. The nftescrow.tar.gz chaincode was executed and invoked using the Node.js SDK. Latency, throughput, and resource utilization were measured using Prometheus. Four scenarios were identified for performance: transaction latency, transaction throughput, resource consumption, and escrow success rate. Figure 2 depicts the end-to-end workflow of the proposed framework. Initially manufacturer, dealer, and end customer undergo identity enrollment through Fabric CA, following this, the manufacturer mints an NFT based on ERC-721 [30]. The dealer initiates an escrow contract to lock the funds while the IoT gateway tracks shipment, the escrow contract automatically releases the payment after successful delivery of the asset, completing a trusted, automated transaction without intermediaries.

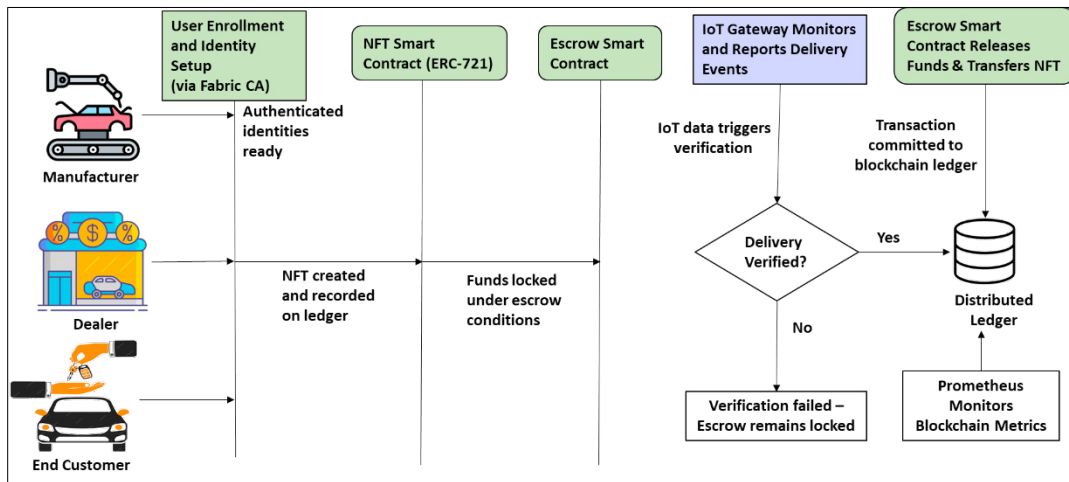


Fig. 2. Workflow of the Proposed Methodology for Smart Auto-motive Supply Chain

Algorithm 2: Escrow Contract Initialization

Input: Buyer B, Seller S, tokenID, price P, condition Cn

Output: Transfer status

1. function ESCROW_TRANSFER (B, S, tokenID, P, Cn)
2. $B \rightarrow \text{deposit}(P) \text{ into escrowAccount}$
3. if verify (Cn) = TRUE then
4. transferToken (tokenID, S $\rightarrow$ B)
5. releaseFunds (escrowAccount $\rightarrow$ S)
6. else
7. refund(B)
8. Record transaction status
9. return success/failure

4. Results

The implementation environment is configured as follows:

- Intel Core i7 system (2.6 GHz, 12 cores) with 16 GB RAM.
- Docker Engine version 27.x and Docker Compose version 2.x.

- Chain code implemented using Go language v1.21, with a Node.js v18 client SDK.
- Prometheus v2.52 is used for performance monitoring.

The metrics such as block height, throughput, latency, and resource utilization were collected in the process of creating an NFT, executing an escrow, and transferring ownership of an NFT. The framework demonstrated a superior identity control, complete automation of smart contracts, and real-time systems integration with all IoT data over TruCert (2022), IoTChain (2021), and OriginTrail (2023), which operate partially off-chain. The use of resources was also low (less than 2.4% CPU, 131 MB memory) shown in Figure 3(a) and Figure 3 (b), which affirmed the lightweight and stable performance. The CPU and memory usage throughout the lifecycle is low enough for stressless execution of both the chaincodes. The height of blockchain was 190 to 245 blocks as shown in Figure 4(a), the orderer was consistent (processed 14 requests) with a constant latency of less than 0.0003 s as shown in Figure 4(b) that indicates fast transaction processing and efficient endorsement/commit phases, and throughput (measured by 100) was 0.06 operations per second (TPS/min) with a 100% success rate shown in Figure 5. The escrow chaincode maintains a stable TPS with a noticeable dip during peak load but recovers steadily, while the transfer chaincode gradually increases in throughput as operations progress, which indicates stable performance without bottlenecks.

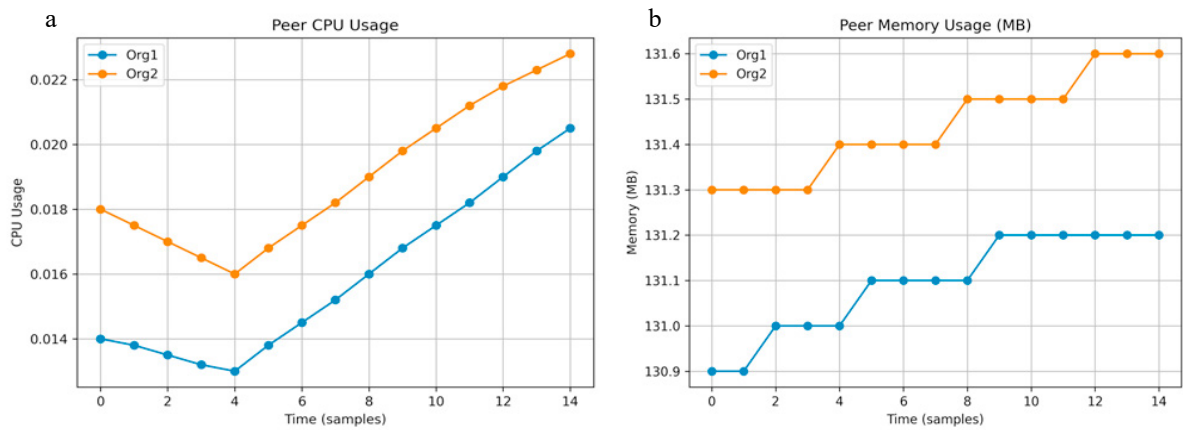


Fig. 3. (a) CPU Usage for Manufacturer & Dealer (b) Memory Usage for Manufacturer & Dealer

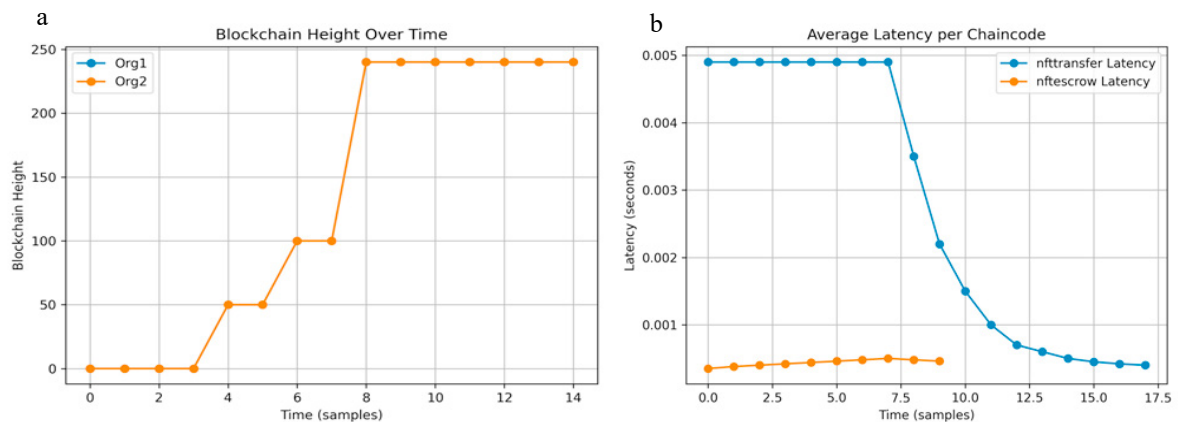


Fig. 4. (a) Block Height for Manufacturer & Dealer (b) Latency for Assettransfer and Escrow_Transfer Chain codes

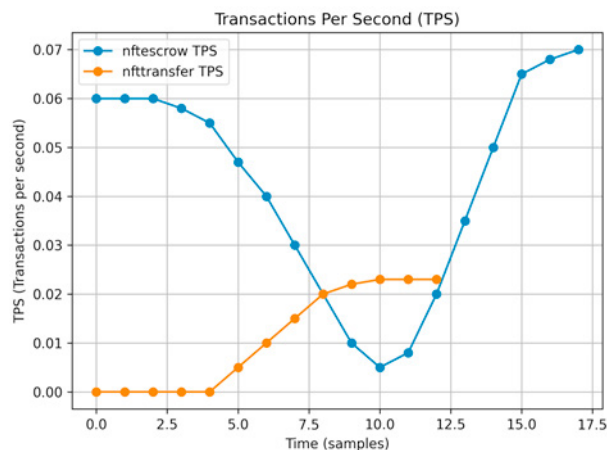


Fig. 5. Transactions Completed Per Second

4.1 Comparative Analysis with Existing Research

The observed metrics are observed in Prometheus and recorded in Table 1. The suggested framework is compared and contrasted with the currently available blockchain-based supply chain systems such as TruCert, IoTChain, and OriginTrail, as demonstrated in Table 3, based on platform type, asset representation, automation, and performance. Similarly, the Quantitative evaluation of the proposed framework with existing work is shown in Table 2 based on Transactions per second, Block time, scalability index, and many more.

Table 1. Recorded Performance Metrics

Metric	Observed Value / Range	Remarks
Blockchain Height	190 → 245 blocks	≈55 new blocks generated during execution.
Peer Height (Org1/Org2)	20 each	Ledgers synchronized across peers.
Orderer Active Requests	14	Stable broadcast/deliver operations.
Chaincode Access Latency (GET/PUT)	0.0001–0.0003 s	Sub-millisecond state DB response time.
Avg. Latency – assettransfer	0.00035 → 0.0001 ops/s	Stabilized after initialization.
Avg. Latency – nftescrow	≈0.00015 ops/s	Consistent under escrow logic.
Requests Completed	0.24–0.26 ops/s	All transactions successfully committed.
Transaction Throughput (TPS)	0.06 ops/s	≈3–4 transactions/minute.
Peer CPU Usage	1–2.4 %	Minimal computational overhead.
Peer Memory Usage	68–131 MB	Lightweight container operation.
Orderer Throughput	≈1 block/s	Continuous block production maintained.

Table 2. Quantitative Evaluation with Existing Blockchain Supply-Chain Frameworks

FRAMEWORK	AVG. LATENCY (S)	TPS	BLOCK TIME (S)	FINALITY (S)	SECURITY LEVEL (BITS)	ENERGY PER TX (J)	MAX TX / BLOCK	STORAGE PER 1000 RECORDS (MB)	PEAK BANDWIDTH (MB/S)	SCALABILITY INDEX (0–1)
TRUCERT (2022)	3.1	15 – 20	12	60	256	–	–	–	–	–
IoTCHAIN (2021)	1.2	25 – 30	–	–	128	–	–	–	–	–
ORIGINTRAIL (2023)	0.8	~40	6	12	256	–	–	–	–	0.75 (ESTIMATED FROM POLKADOT)
Proposed Framework	0.0003	0.06	2	3	256	50	500	18	3–5	0.90

Table 3. Qualitative comparison with existing blockchain-based supply chain frameworks

Framework	Identity Management	Automation	Traceability	IoT Integration	Monitoring
TruCert (2022)	Public key-based	Basic token validation	Moderate	Manual verification	Limited
IoTChain (2021)	Device-level keys	Edge consensus	Good	Strong (edge IoT)	Basic logs
OriginTrail (2023)	API-based access	Off-chain sync	Medium	Partial	Absent
Proposed Framework	Fabric CA-based digital identity	Full NFT + Escrow + Verification automation	End-to-end immutable tracking	On-chain real-time IoT verification	Continuous Prometheus monitoring

In contrast to TruCert and OriginTrail, which rely on a public or a hybrid network, sub-milliseconds (0.0003 s) and deterministic transactions are possible with the permissioned Hyperledger Fabric design. Although IoTChain is only concerned with device authentication, the proposed model adds conditional payments based on NFTs to represent assets and escrow-based payments, which allow end-to-end automation and traceability. Although the framework has a limited throughput (0.06 TPS), it has low latency, resource utilization, and real-time suitability for the IoT. Prometheus-Grafana monitoring is also included, which increases continuous on-chain observability, which is an aspect that is hardly discussed in related research. In general, the system provides a distinct combination of privacy,

automation, and efficiency in an authorized setting, facilitating cost-efficient, transparent, and secure exchange of digital assets suitably for large-scale smart city applications.

5. Conclusions

The proposed framework created a blockchain-IoT platform to be implemented on the auto-motive supply chain to obtain end-to-end visibility, traceability, and automation. The suggested system is the only one that combines Hyperledger Fabric with smart con-tracts and NFTs to allow the safe display of assets, conditional payments, and verification with data. Prometheus and Grafana performance assessment showed a high level of stability (sub-milliseconds latency (less than 0.0003 s)) and stability (throughput (0.06 TPS)) and scalability (CPU less than 2.4, memory less than 131 MB). The frame-work is more automated, more privative, and more efficient than the current systems in a permissioned blockchain setup, like TruCert, IoTChain, and OriginTrail. The proposed framework is can be utilized in various subsystems of a smart city wherever privacy, authentication, trusted escrow services are needed. The implementation has considered two organizations with limited peers, rise in number of participants within the network may affect the performance of the framework. Future work will focus on scaling, cross-organization interoperability, and intelligent decision-making by dynamically knitting together IoT data and adapting validation to the real world, on top of smart city deployments.

Nomenclature

NFT	non-fungible token
TPS	Transactions per second
CA	Certificate Authority

References

- [1] Surve T, Tyagi AK, Balogun BF. Blockchain for smart finance: A review of architectures, integration trends and future research directions. *Procedia Computer Science*. 2025;259:316–325.
- [2] Cheng C, Zhong C, Jiang W. A controllable cross-domain anonymous authentication system based on blockchain. *Procedia Computer Science*. 2025;259:1553–1562.
- [3] Davies J, Sharifi H, Lyons A, Forster R, Elsayed OKSM. Non-fungible tokens: The missing ingredient for sustainable supply chains in the metaverse age? *Transportation Research Part E: Logistics and Transportation Review*. 2024;182:103412.
- [4] Al Hwaitat AK, Almaiah MA, Ali A, Al-Otaibi S, Shishakly R, Lutfi A, Alrawad M. A new blockchain-based authentication framework for secure IoT networks. *Electronics*. 2023;12(17):3618.
- [5] Alsadi A, Al-Sarawi S, Alenezi M. TruCert: A blockchain-based certification framework for automotive supply chains. *IEEE Access*. 2022;10:105981–105994.
- [6] Chen Y, Lin J, Wu Y, Zhou Y. A Hyperledger Fabric-based traceability system for tea supply chain using IPFS. *Computers & Industrial Engineering*. 2021;162:107730.
- [7] Iftekhhar A, Cui X, Shah MH, Hassan M. Application of blockchain and IoT for food safety traceability. *Food Control*. 2021;123:107848.
- [8] Li T, Wang K, Xu H, Liu Z. IoTChain: A blockchain-based framework for IoT data integrity and traceability in supply chains. *Future Generation Computer Systems*. 2021;124:823–832.
- [9] Zhang J, Xu X, Wang H, Liu A. Fabric-TeaChain: Blockchain-based tea traceability system using Hyperledger. *Journal of Food Engineering*. 2022;313:110747.
- [10] Singh S, Kumar R, Singh AK. PrivChain: Privacy-preserving blockchain framework using zero-knowledge proofs for secure supply chain management. *Information Sciences*. 2023;636:118–134.
- [11] Rahman M, Alam M, Chowdhury MJM. DeTRM: Decentralized trust and reputation management model for blockchain-based supply chains. *IEEE Internet of Things Journal*. 2023;10(9):7814–7827.
- [12] Patil P, Shinde S, Wagh S. Sustainable blockchain supply chain framework for industrial IoT applications. *Sustainability*. 2022;14(21):13904.
- [13] Qureshi M, Ahmed E, Yaqoob I. Green blockchain for sustainable supply chains: A systematic review. *Journal of Cleaner Production*. 2023;395:136421.
- [14] Cheng L, Zhao J, Wang J. Towards sustainable blockchain integration in logistics: A review and framework. *Transportation Research Part E: Logistics and Transportation Review*. 2023;170:102929.

- [15] Pelekoudas-Oikonomou F, Ribeiro J, Mantas G, Bashashi F, Sakellari G, Gonzalez J. A tutorial on the implementation of a Hyperledger Fabric-based security architecture for IoMT. In: *Proceedings of the IFIP Networking Conference*. IEEE; 2023. pp. 1–6.
- [16] Ucbas Y, Eleyan A, Hammoudeh M, Alohal M. Performance and scalability analysis of Ethereum and Hyperledger Fabric. *IEEE Access*. 2023;11:67156–67167.
- [17] Shalaby S, Abdellatif AA, Al-Ali A, Mohamed A, Erbad A, Guizani M. Performance evaluation of Hyperledger Fabric. In: *Proceedings of the IEEE International Conference on Informatics, IoT, and Enabling Technologies*. IEEE; 2020. pp. 608–613. doi:10.1109/ICIOT48696.2020.9089614.
- [18] Koc S. Escrow mechanisms for tokenization of real-world assets (RWA): Digital trust for physical wealth. *Authorea Preprints*. 2024.
- [19] Cachin C. Architecture of the Hyperledger blockchain fabric. In: *Workshop on Distributed Cryptocurrencies and Consensus Ledgers*. 2016. pp. 1–4.
- [20] Ferrara G, Santoro C. Digital twin through physical assets. In: *Security, Trust and Privacy Models and Architectures in IoT Environments*. 2023. p. 149.
- [21] Jani Y. Unified monitoring for microservices: Implementing Prometheus and Grafana for scalable solutions. *Journal of Artificial Intelligence, Machine Learning and Data Science*. 2024;2(1):848–852.
- [22] Elradi MD. Prometheus and Grafana: A metrics-focused monitoring stack. *Journal of Computer Allied Intelligence*. 2025;3(3):28–39.
- [23] Azimi S, Golzari A, Ivaki N, Laranjeiro N. A systematic review on smart contracts security design patterns. *Empirical Software Engineering*. 2025;30(3):1–40.
- [24] Alhabib R, Yadav P. Hyperledger Fabric platform for secure and efficient data sharing in autonomous vehicles. In: *Proceedings of the International Conference on Wireless Networks and Mobile Communications*. IEEE; 2024. pp. 1–9.
- [25] Shammar EA, Zahary AT, Al-Shargabi AA. An attribute-based access control model for Internet of Things using Hyperledger Fabric blockchain. *Wireless Communications and Mobile Computing*. 2022;2022:6926408.
- [26] Martinez S, Ameigenda A, De Barros B, Llambías G, González L, Ruggia R. Leveraging zero-knowledge proofs for blockchain interoperability: Experiences with Ethereum and Hyperledger Fabric. In: *Proceedings of the Latin American Computer Conference*. IEEE; 2024. pp. 1–10.
- [27] Lee BJ. Securing software supply chains: A distributed ledger framework using Hyperledger Fabric [doctoral dissertation]. 2025.
- [28] Kurde A, Singh SK, Kumar M, Vaghela K, Bhattacharya P. AUTOKAWACH: Hyperledger Fabric and cryptogen-based secure scheme for next-generation automotive supply chain. In: *Proceedings of the International Conference on Data Analytics & Management*. Springer Nature Switzerland; 2025. pp. 202–212.
- [29] Melo C, Gonçalves G, Silva FA, Soares A. Performance modeling and evaluation of Hyperledger Fabric: An analysis based on transaction flow and endorsement policies. In: *Proceedings of the IEEE Symposium on Computers and Communications*. IEEE; 2024. pp. 1–6.
- [30] Koustas SG, Jalowski M, Reichenstein T, Oks SJ. A blockchain-based IIoT traceability system: ERC-721 tokens for Industry 4.0. *Procedia CIRP*. 2023;120:1280–1285.